HDOC DAY 66

PROBLEM

Write a recursive program to **multiply two square matrices**. Do not use iterative loops for the multiplication logic.

CODE

```
#include <stdio.h>


#define N 3


void multiply(int A[N][N], int B[N][N], int C[N][N],
          int row, int col, int k)
{
   // All rows completed
   if (row == N)
      return;


   // Current row completed
   if (col == N)
   {
      multiply(A, B, C, row + 1, 0, 0);
      return;
   }


   // Calculate C[row][col]
   if (k < N)
   {
```

```
        C[row][col] += A[row][k] * B[k][col];
        multiply(A, B, C, row, col, k + 1);
    }
    else
    {
        multiply(A, B, C, row, col + 1, 0);
    }
}

int main()
{
    int A[N][N] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };

    int B[N][N] = {
        {9, 8, 7},
        {6, 5, 4},
        {3, 2, 1}
    };

    int C[N][N] = {0};

    multiply(A, B, C, 0, 0, 0);

    printf("Product Matrix:\n");
```

```
    for (int i = 0; i < N; i++)
    {
        for (int j = 0; j < N; j++)
            printf("%d ", C[i][j]);

        printf("\n");
    }

    return 0;
}
```